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DSA0602 - DATA HANDLING AND VISUALIZATION
(R Programming Edition)

1. Student Academic Performance Dashboard

Learning Outcome

Understand how position and color aesthetics represent different data types.

Scenario

The Head of the Computer Science Department wants to analyze whether students with higher attendance perform better in examinations. The department also wants to compare the performance across different programs.

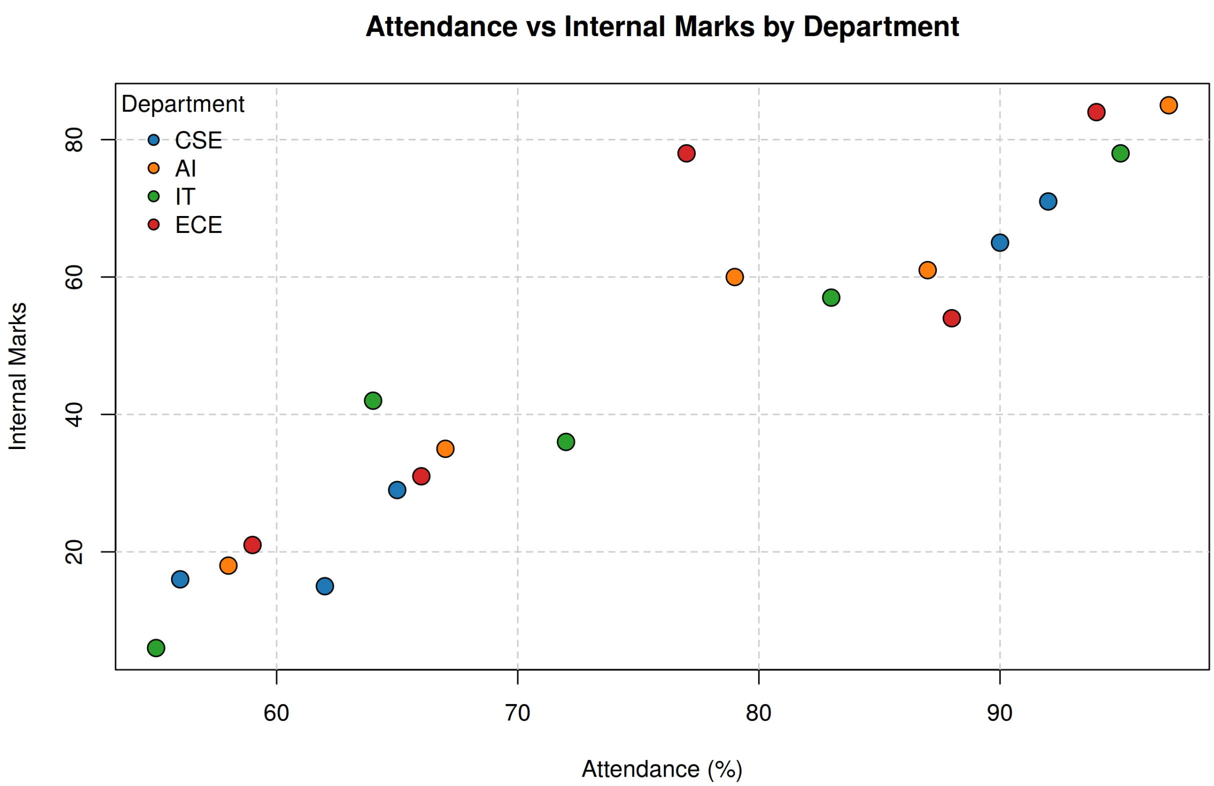
Dataset

- Student ID
- Attendance (%)
- Internal Marks
- Department (CSE, AI, IT, ECE)

R Code

```
# Scenario 1: Student Academic Performance Dashboard

dept <- c("CSE","CSE","CSE","CSE","CSE","AI","AI","AI","AI","AI",
          "IT","IT","IT","IT","IT","ECE","ECE","ECE","ECE","ECE")
attendance <- c(56,62,65,90,92, 58,67,79,87,97, 55,64,72,83,95, 59,66,77,88,94)
marks      <- c(16,15,29,65,71, 18,35,60,61,85, 6,42,36,57,78,  21,31,78,54,84)

colors <- c(CSE="#1f77b4", AI="#ff7f0e", IT="#2ca02c", ECE="#d62728")

plot(attendance, marks, type="n",
     main="Attendance vs Internal Marks by Department",
     xlab="Attendance (%)", ylab="Internal Marks")
grid(lty=2, col="gray80")

for (d in names(colors)) {
  idx <- dept == d
  points(attendance[idx], marks[idx], pch=21, bg=colors[d], col="black", cex=1.6)
}

legend("topleft", legend=names(colors), pt.bg=colors, pch=21,
       title="Department", bty="n")
```

Output



Attendance vs Internal Marks by Department
Department
CSE
AI
IT
ECE
Internal Marks
Attendance (%)
20
40
60
80
60
70
80
90

2. Electric Vehicle Market Analysis

Learning Outcome

Learn how size and color aesthetics can simultaneously represent multiple variables.

Scenario

An automobile company is studying the relationship between battery capacity and vehicle price to understand whether larger batteries significantly increase the selling price.

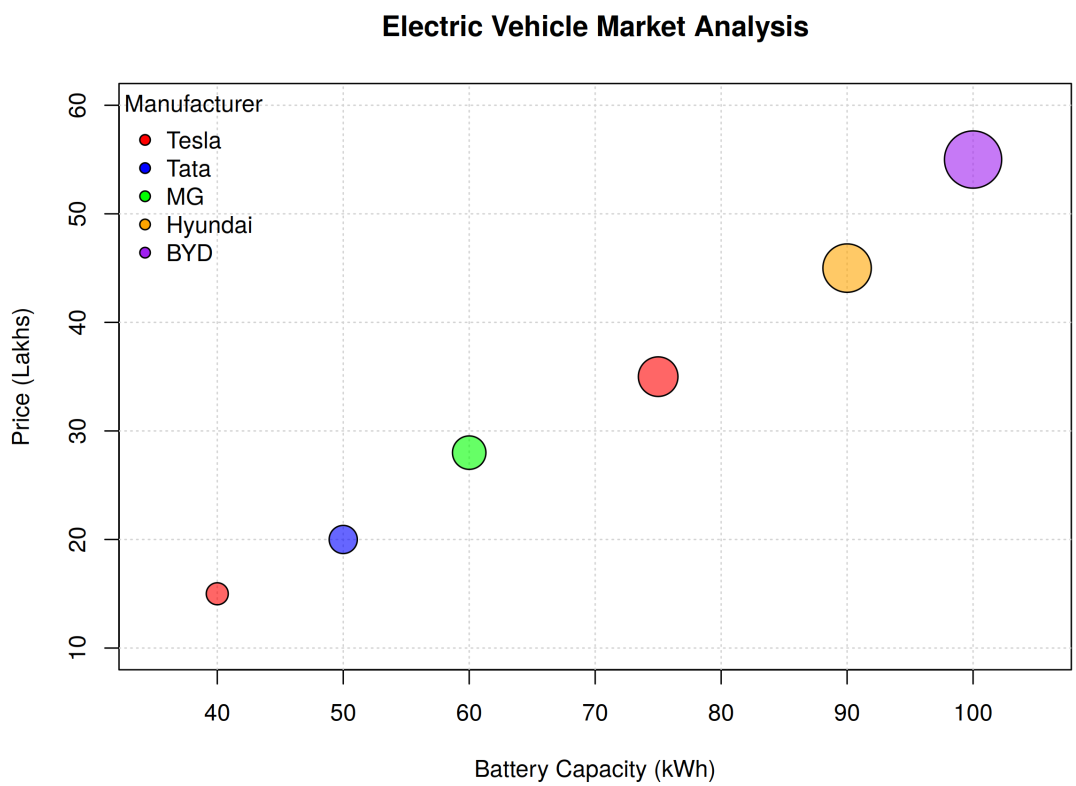
Dataset

- Vehicle Model
- Battery Capacity (kWh)
- Price
- Driving Range
- Manufacturer

R Code

```
# Scenario 2: Electric Vehicle Market Analysis

battery <- c(40, 50, 60, 75, 90, 100)
price <- c(15, 20, 28, 35, 45, 55)
range_km <- c(250, 320, 380, 450, 550, 650)
manufacturer <- c("Tesla","Tata","MG","Tesla","Hyundai","BYD")

colors <- c(Tesla="red", Tata="blue", MG="green", Hyundai="orange", BYD="purple")

plot(battery, price, type="n",
     main="Electric Vehicle Market Analysis",
     xlab="Battery Capacity (kWh)", ylab="Price (Lakhs)",
     xlim=c(35,105), ylim=c(10,60))
grid()

# Bubble size represents driving range
cex_vals <- range_km / 120
points(battery, price, pch=21,
       bg=adjustcolor(colors[manufacturer], alpha.f=0.6),
       col="black", cex=cex_vals)

legend("topleft", legend=names(colors), pt.bg=colors, pch=21,
       title="Manufacturer", bty="n")
```

Output


Electric Vehicle Market Analysis
Manufacturer
Tesla
Tata
MG
Hyundai
BYD
Price (Lakhs)
10
20
30
40
50
60
40
50
60
70
80
90
100
Battery Capacity (kWh)

3. Hospital Patient Admission Analysis

Learning Outcome

Visualize categorical variables using fill color and position aesthetics.

Scenario

A hospital administrator wants to compare patient admissions across departments during the last month.

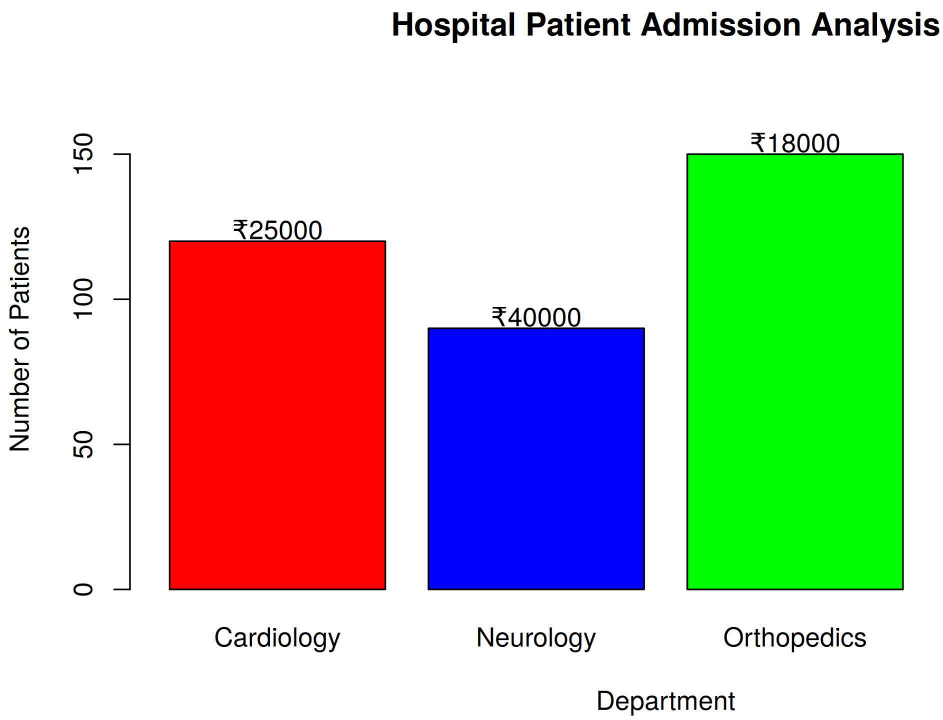
Dataset

- Department
- Number of Patients
- Average Treatment Cost

R Code

```r
# Scenario 3: Hospital Patient Admission Analysis

departments <- c("Cardiology","Neurology","Orthopedics","Pediatrics")
patients <- c(120, 90, 150, 110)
cost <- c(25000, 40000, 18000, 12000)
colors <- c("red","blue","green","orange")

bar_mid <- barplot(patients, names.arg=departments, col=colors,
                    main="Hospital Patient Admission Analysis",
                    xlab="Department", ylab="Number of Patients",
                    ylim=c(0, max(patients)*1.15))

# Display treatment cost above each bar
text(bar_mid, patients + 4, labels=paste0("\u20b9", cost))
```

Output


Hospital Patient Admission Analysis
Number of Patients
0
50
100
150
₹25000
₹40000
₹18000
₹12000
Cardiology
Neurology
Orthopedics
Pediatrics
Department

4. Weather Monitoring System

Learning Outcome

Understand how line type and color distinguish multiple categories.

Scenario

The Indian Meteorological Department has collected daily temperature data for three cities over one month and wants to compare climate trends.

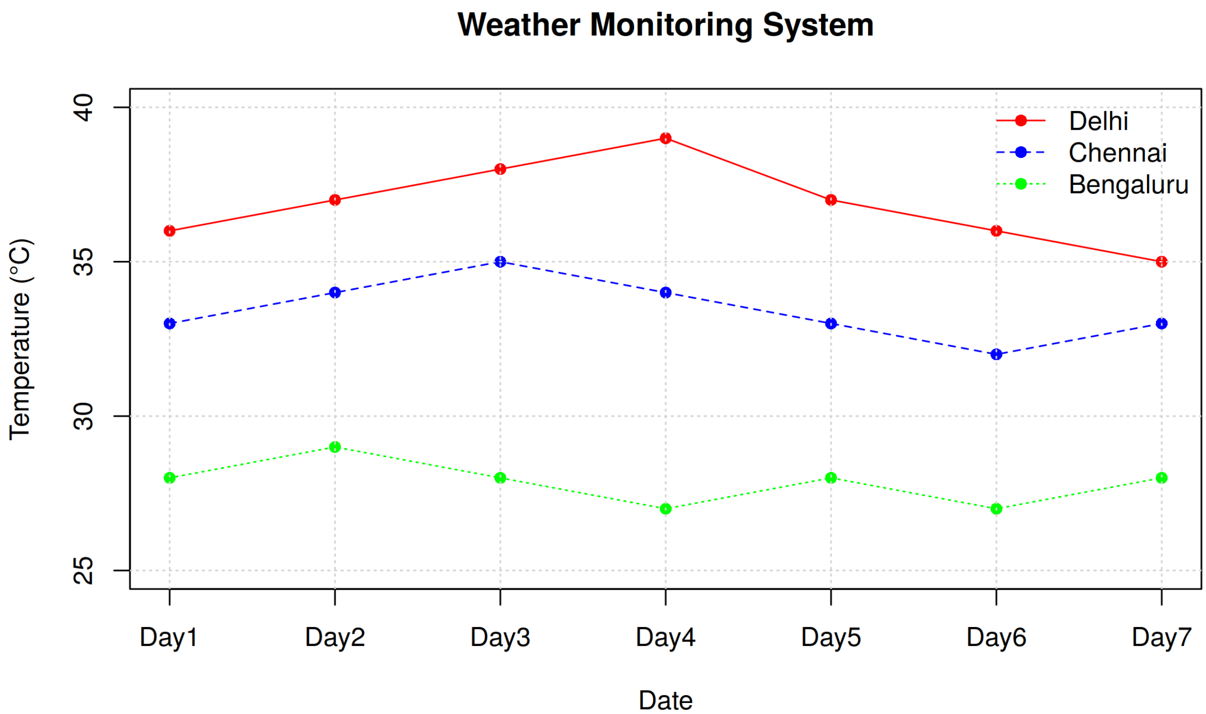
Dataset

- Date
- Temperature
- City

R Code

```
# Scenario 4: Weather Monitoring System

dates <- paste0("Day", 1:7)
delhi <- c(36, 37, 38, 39, 37, 36, 35)
chennai <- c(33, 34, 35, 34, 33, 32, 33)
bengaluru <- c(28, 29, 28, 27, 28, 27, 28)

plot(1:7, delhi, type="o", col="red", lty=1, pch=16, ylim=c(25,40),
     xaxt="n", main="Weather Monitoring System",
     xlab="Date", ylab="Temperature (\u00b0C)")
axis(1, at=1:7, labels=dates)

lines(1:7, chennai, type="o", col="blue", lty=2, pch=16)
lines(1:7, bengaluru, type="o", col="green", lty=3, pch=16)
grid()

legend("topright", legend=c("Delhi","Chennai","Bengaluru"),
       col=c("red","blue","green"), lty=c(1,2,3), pch=16, bty="n")
```

Output


Weather Monitoring System
Delhi
Chennai
Bengaluru
Temperature (°C)
40
35
30
25
Day1
Day2
Day3
Day4
Day5
Day6
Day7
Date

5. Supermarket Sales Analysis

Learning Outcome

Apply position and fill aesthetics for grouped comparisons.

Scenario

A supermarket chain wants to identify which product categories generate the highest monthly sales across different branches.

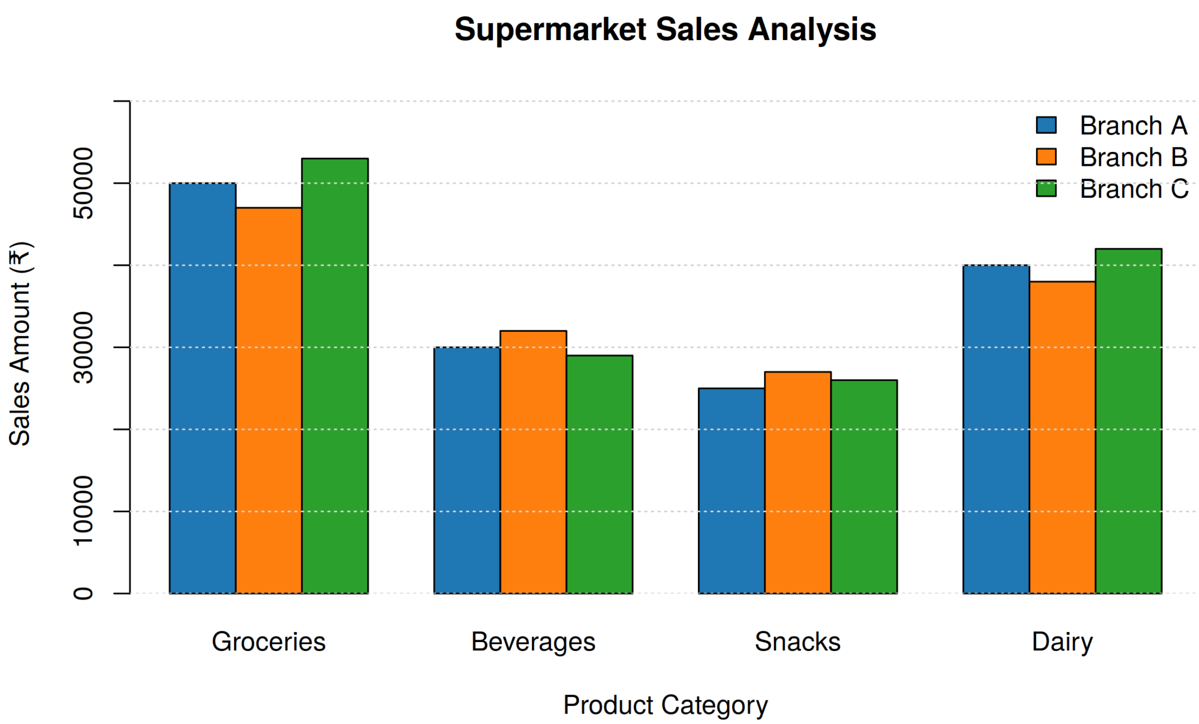
Dataset

- Month
- Product Category
- Sales Amount
- Branch

R Code

```
# Scenario 5: Supermarket Sales Analysis

categories <- c("Groceries","Beverages","Snacks","Dairy")
branchA <- c(50000, 30000, 25000, 40000)
branchB <- c(47000, 32000, 27000, 38000)
branchC <- c(53000, 29000, 26000, 42000)

mat <- rbind(branchA, branchB, branchC)
colnames(mat) <- categories

barplot(mat, beside=TRUE, col=c("#1f77b4","#ff7f0e","#2ca02c"),
        main="Supermarket Sales Analysis",
        xlab="Product Category", ylab="Sales Amount (\u20b9)",
        ylim=c(0, max(mat)*1.15))

legend("topright", legend=c("Branch A","Branch B","Branch C"),
       fill=c("#1f77b4","#ff7f0e","#2ca02c"), bty="n")
grid(nx=NA, ny=NULL)
```

Output



Supermarket Sales Analysis
Branch A
Branch B
Branch C
Sales Amount (₹)
0
10000
30000
50000
Groceries
Beverages
Snacks
Dairy
Product Category

6. Employee Salary Distribution

Learning Outcome

Understand continuous scales and histogram binning.

Scenario

The HR department wants to understand salary distribution among employees to determine whether salaries are evenly distributed across the organization.

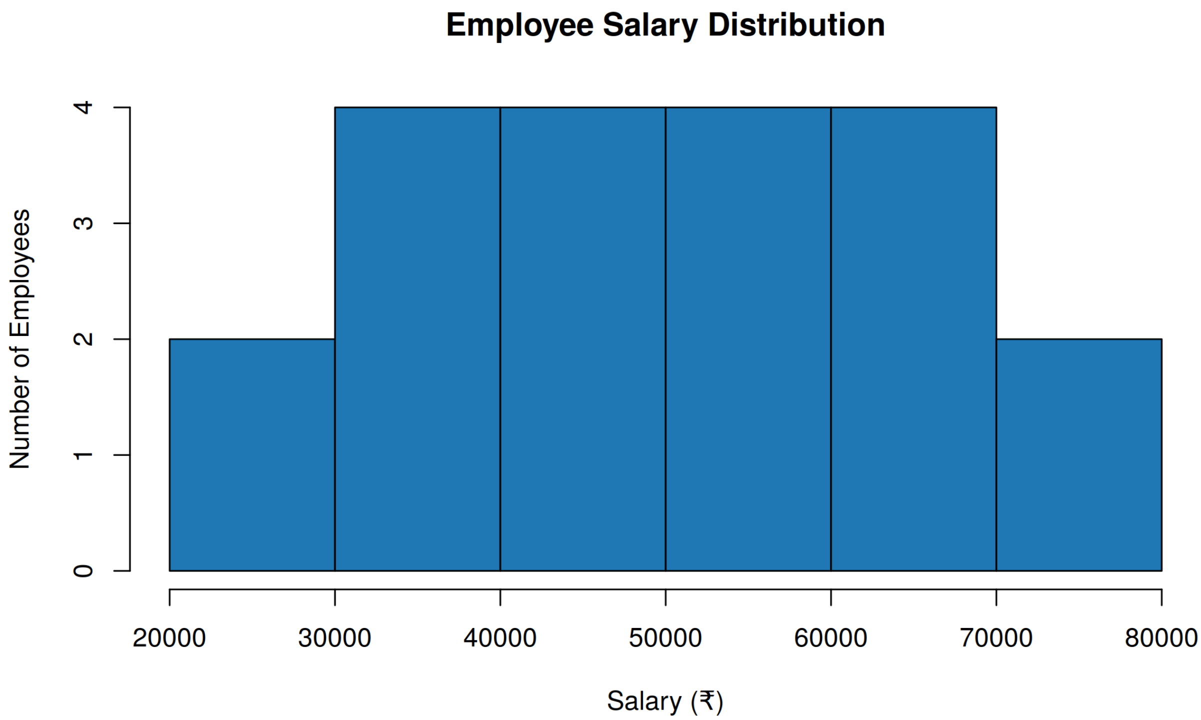
Dataset

- Employee ID
- Salary
- Experience
- Department

R Code

```
# Scenario 6: Employee Salary Distribution

salary <- c(25000,30000,32000,35000,38000,40000,42000,45000,
            47000,50000,52000,55000,58000,60000,62000,
            65000,68000,70000,75000,80000)

# Histogram with 5 bins
hist(salary, breaks=5, col="#1f77b4", border="black",
     main="Employee Salary Distribution",
     xlab="Salary (\u20b9)", ylab="Number of Employees")
```

Output


Employee Salary Distribution
Number of Employees
0
1
2
3
4
20000
30000
40000
50000
60000
70000
80000
Salary (₹)

7. Smartphone Market Comparison

Learning Outcome

Map multiple variables using shape, size, and color.

Scenario

A technology company wants to compare smartphone specifications to understand how RAM and processor speed influence product pricing.

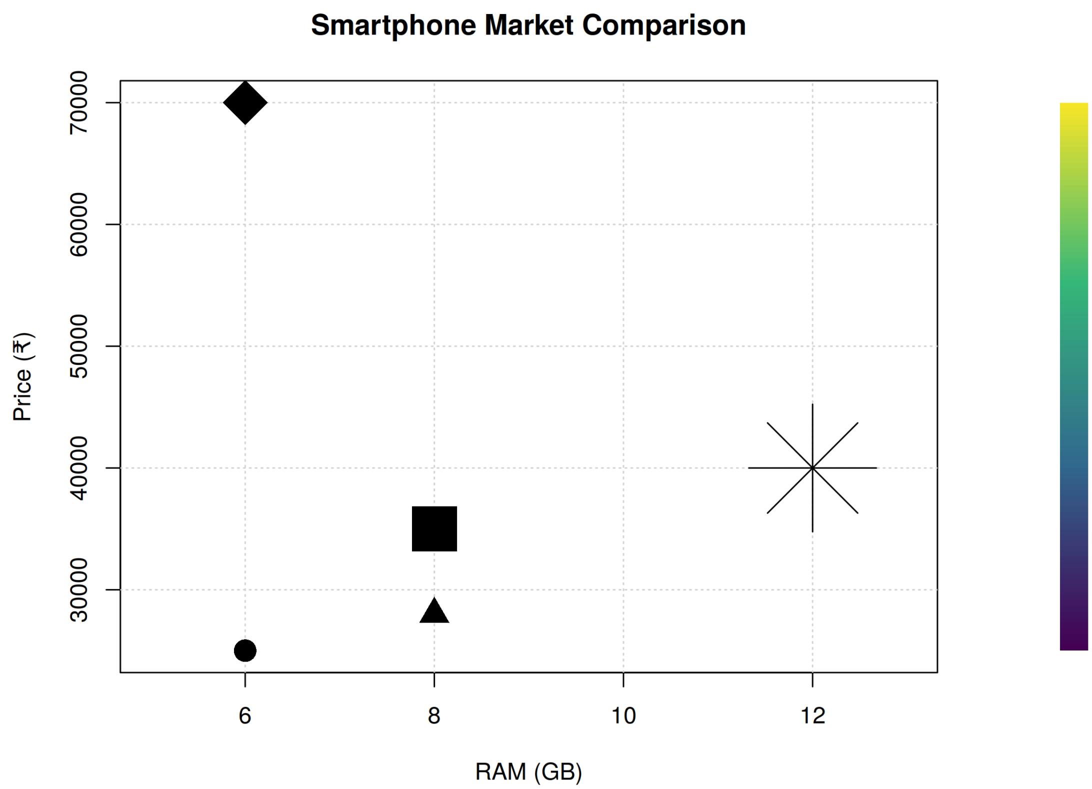
Dataset

- Brand
- RAM
- Processor Speed
- Price
- Storage

R Code

```
# Scenario 7: Smartphone Market Comparison

ram <- c(6, 8, 8, 6, 12)
price <- c(25000, 35000, 28000, 70000, 40000)
storage <- c(128, 256, 128, 256, 512)
processor <- c(2.2, 2.6, 2.4, 3.2, 2.8)
brands <- c("Samsung","OnePlus","Xiaomi","Apple","Realme")
pch_vals <- c(16, 15, 17, 18, 8)   # marker shape per brand

# Continuous color scale (viridis-like) for processor speed
ramp <- colorRamp(c("#440154","#31688e","#35b779","#fde725"))
proc_norm <- (processor - min(processor)) / (max(processor) - min(processor))
cols <- apply(ramp(proc_norm), 1,
              function(rgb) rgb(rgb[1], rgb[2], rgb[3], maxColorValue=255))

plot(ram, price, type="n", main="Smartphone Market Comparison",
     xlab="RAM (GB)", ylab="Price (\u20b9)", xlim=c(5,13))
grid()

# Point size represents storage
points(ram, price, pch=pch_vals, col="black", bg=cols, cex=storage/60)

# (color legend strip for Processor Speed drawn alongside the plot)
```

Output



Smartphone Market Comparison
Price (₹)
70000
60000
50000
40000
30000
6
8
10
12
RAM (GB)

8. Online Movie Rating Analysis

Learning Outcome

Understand continuous color scales and gradient aesthetics.

Scenario

A movie streaming platform wants to compare average ratings received by different movie genres over the past year.

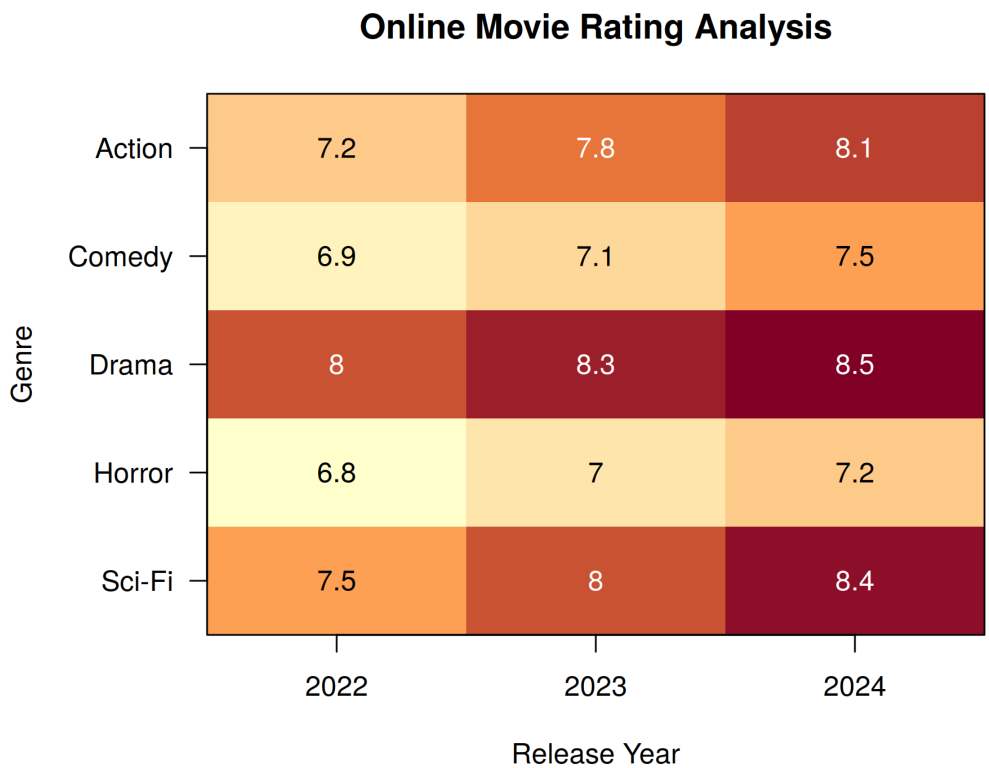
Dataset

- Genre
- Average Rating
- Number of Reviews
- Release Year

R Code

```
# Scenario 8: Online Movie Rating Analysis

genres <- c("Action","Comedy","Drama","Horror","Sci-Fi")
years <- c("2022","2023","2024")

mat <- matrix(c(7.2,7.8,8.1,
                6.9,7.1,7.5,
                8.0,8.3,8.5,
                6.8,7.0,7.2,
                7.5,8.0,8.4),
              nrow=5, byrow=TRUE,
              dimnames=list(genres, years))

heat_cols <- colorRampPalette(c("#ffffcc","#fd8d3c","#800026"))(100)
mat_flip <- mat[nrow(mat):1, ]

image(1:ncol(mat_flip), 1:nrow(mat_flip), t(mat_flip), col=heat_cols,
      axes=FALSE, xlab="Release Year", ylab="",
      main="Online Movie Rating Analysis")
axis(1, at=1:ncol(mat_flip), labels=colnames(mat_flip))
axis(2, at=1:nrow(mat_flip), labels=rownames(mat_flip), las=1)
mtext("Genre", side=2, line=5)

for (i in 1:nrow(mat_flip))
  for (j in 1:ncol(mat_flip))
    text(j, i, mat_flip[i,j],
         col=ifelse(mat_flip[i,j] > 7.7, "white", "black"))
box()
```

Output


Online Movie Rating Analysis
Genre
Action
Comedy
Drama
Horror
Sci-Fi
7.2
7.8
8.1
6.9
7.1
7.5
8
8.3
8.5
6.8
7
7.2
7.5
8
8.4
2022
2023
2024
Release Year

9. COVID-19 State-wise Analysis

Learning Outcome

Understand sequential, diverging, and categorical color scales.

Scenario

The Ministry of Health wants to visualize the distribution of COVID-19 cases across different Indian states to identify highly affected regions.

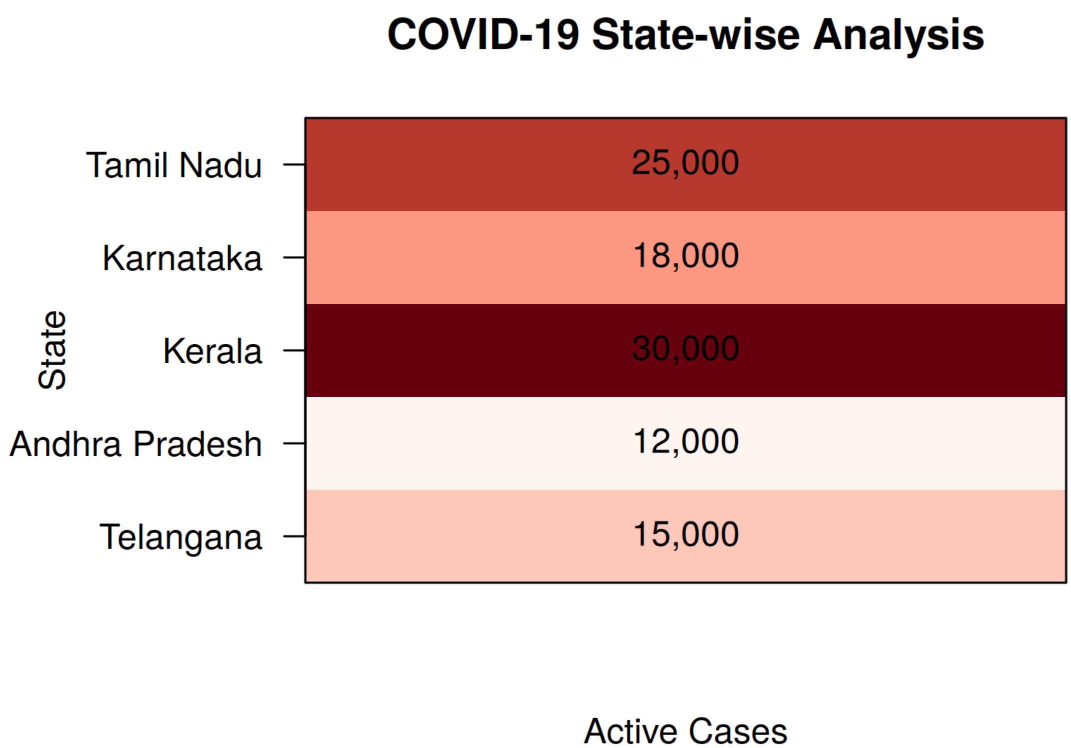
Dataset

- State
- Active Cases
- Vaccination Percentage
- Population

R Code

```
# Scenario 9: COVID-19 State-wise Analysis

states <- c("Tamil Nadu","Karnataka","Kerala","Andhra Pradesh","Telangana")
active_cases <- c(25000, 18000, 30000, 12000, 15000)

mat <- matrix(active_cases, ncol=1, dimnames=list(states, "Active Cases"))
mat_flip <- mat[nrow(mat):1, , drop=FALSE]

heat_cols <- colorRampPalette(c("#fff5f0","#fb6a4a","#67000d"))(100)

image(1, 1:nrow(mat_flip), t(mat_flip), col=heat_cols, axes=FALSE,
      xlab="Active Cases", ylab="",
      main="COVID-19 State-wise Analysis")
axis(2, at=1:nrow(mat_flip), labels=rownames(mat_flip), las=1)
mtext("State", side=2, line=5.5)

for (i in 1:nrow(mat_flip))
  text(1, i, format(mat_flip[i,1], big.mark=","), col="black")
box()
```

Output


COVID-19 State-wise Analysis
Tamil Nadu
Karnataka
Kerala
Andhra Pradesh
Telangana
State
25,000
18,000
30,000
12,000
15,000
Active Cases

10. Iris Flower Classification

Learning Outcome

Understand how multiple aesthetic mappings improve pattern recognition and class separation.

Scenario

A botanist wants to analyze flower measurements to determine whether different Iris species can be visually separated.

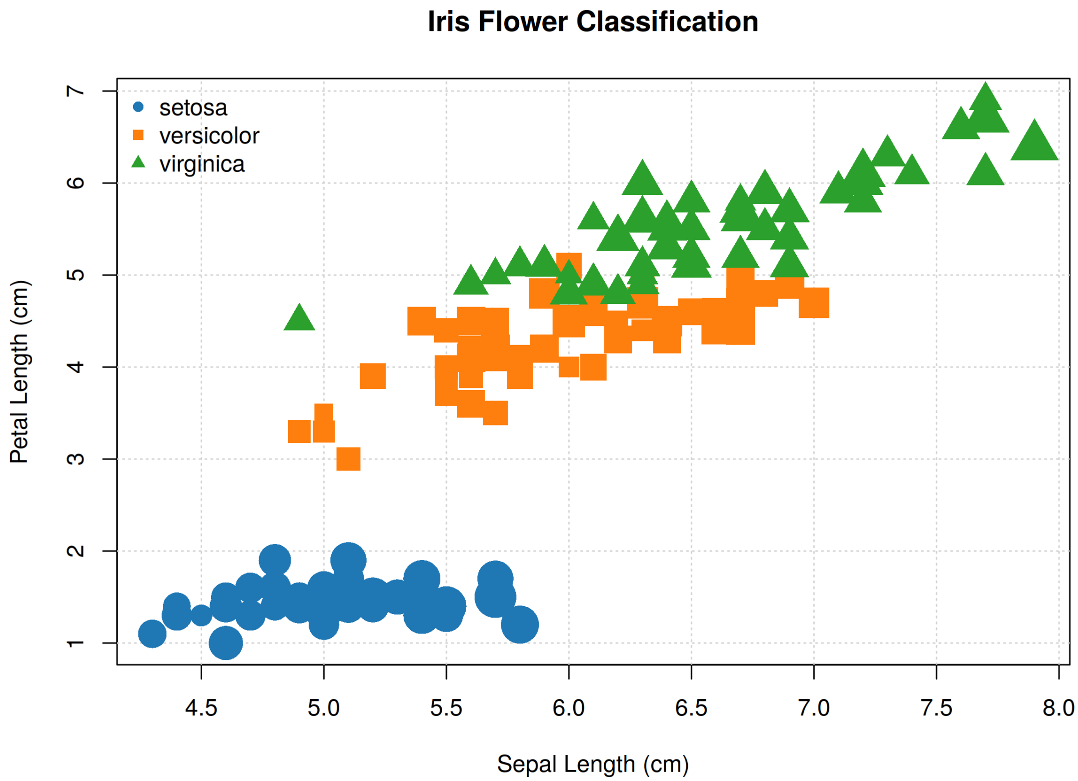
Dataset

- (Built-in Iris Dataset)
- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species

R Code

```
# Scenario 10: Iris Flower Classification

data(iris)  # built-in dataset

species <- iris$Species
levels_sp <- levels(species)
pch_vals <- c(16, 15, 17)
cols <- c("#1f77b4", "#ff7f0e", "#2ca02c")

plot(iris$Sepal.Length, iris$Petal.Length, type="n",
     main="Iris Flower Classification",
     xlab="Sepal Length (cm)", ylab="Petal Length (cm)")
grid()

for (i in 1:3) {
  idx <- species == levels_sp[i]
  points(iris$Sepal.Length[idx], iris$Petal.Length[idx],
         pch=pch_vals[i], col=cols[i],
         cex=iris$Sepal.Width[idx] * 0.9)  # Sepal Width as point size
}

legend("topleft", legend=levels_sp, pch=pch_vals, col=cols, bty="n")
```

Output



Iris Flower Classification
setosa
versicolor
virginica
Petal Length (cm)
Sepal Length (cm)
1
2
3
4
5
6
7
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0